

SQL and Database Fundamentals

Lab 2

Description: Using the School dataset, answer the following 20 questions which focus on DQL (SELECT), filtering, Joins, Grouping, Views, Ranking Functions, CTE, ORDER BY, and TOP

- 1) Retrieve number of students who have a value in their age
- 2) Get all instructor names without repetition (DISTINCT)
- 3) Display student with the following format: Student ID, Student Full Name, Department Name using ISNULL
- 4) Display instructor name and department name. Display all instructors whether attached to a department or not
- 5) Display student full name and the course name for only courses which have a grade
- 6) Display number of courses for each topic name
- 7) Display maximum and minimum salary for instructors
- 8) Display instructors who have salaries less than the average salary of all instructors
- 9) Display the department name that contains the instructor who receives the minimum salary

- 10) Select instructor name and salary, but if salary is NULL display 'Bonus' using COALESCE function
- 11) Display instructors ordered by salary descending (ORDER BY)
- 12) Display the top 5 highest paid instructors (TOP with ORDER BY)
- 13) Display the top 3 departments with the most students (TOP and ORDER BY)
- 14) Create an encrypted view that displays manager names and the topics they teach
- 15) Create a view that displays instructor name and department name for the 'SD' or 'Java' department
- 16) Create a view "V1" that displays student data for students who live in Alex or Cairo
- 17) Create a view that displays student full name and course name if the student has a grade more than 50
- 18) Write a query using CTE to display department name with the maximum number of students
- 19) Write a query using CTE to display the second highest salary instructor in each department
- 20) Write a query using CTE to display the top 2 highest-paid instructors in each department